

## Exponential functions



- 1      a Yes      b No
- 2   **A**
- 3      a Decreasing      b Increasing      c Decreasing      d Increasing  
     e Increasing      f Increasing      g Decreasing      h Decreasing  
     i Decreasing      j Increasing      k Increasing      l Increasing
- 4   **B**
- 5      a  $y = 5^x$       b  $y = 3 \times 2^x$
- 6      a 1      b -2      c 3      d 3
- 7      a -243      b -3      c  $-\frac{1}{3}$       d 4
- 8   67
- 9      a 1      b  $\frac{9}{16}$
- 10 50 313.28
- 11 6885.29
- 12 3313.46

13



a

i  $\frac{1}{16}$

ii 1

iii 16

- b None. For all values of  $x$ , it will have a positive value.
- c It gets larger, approaching infinity.
- d It gets smaller, approaching zero.

14

a

|     |                 |                |                |               |               |   |   |   |    |    |     |        |
|-----|-----------------|----------------|----------------|---------------|---------------|---|---|---|----|----|-----|--------|
| $x$ | -5              | -4             | -3             | -2            | -1            | 0 | 1 | 2 | 3  | 4  | 5   | 10     |
| $y$ | $\frac{1}{243}$ | $\frac{1}{81}$ | $\frac{1}{27}$ | $\frac{1}{9}$ | $\frac{1}{3}$ | 1 | 3 | 9 | 27 | 81 | 243 | 59 049 |

- b As  $x$  increases, the function increases at a faster and faster rate.
- c All real  $x$
- d  $y > 0$

15

a

|     |    |    |    |    |    |   |               |               |               |                |                |                  |
|-----|----|----|----|----|----|---|---------------|---------------|---------------|----------------|----------------|------------------|
| $x$ | -5 | -4 | -3 | -2 | -1 | 0 | 1             | 2             | 3             | 4              | 5              | 10               |
| $y$ | 32 | 16 | 8  | 4  | 2  | 1 | $\frac{1}{2}$ | $\frac{1}{4}$ | $\frac{1}{8}$ | $\frac{1}{16}$ | $\frac{1}{32}$ | $\frac{1}{1024}$ |

- b As  $x$  increases, the function decreases at a slower and slower rate.
- c 1
- d All real  $x$
- e  $y > 0$

16

a

|     |                 |                 |                |                |                |    |    |    |    |     |     |       |
|-----|-----------------|-----------------|----------------|----------------|----------------|----|----|----|----|-----|-----|-------|
| $x$ | -5              | -4              | -3             | -2             | -1             | 0  | 1  | 2  | 3  | 4   | 5   | 10    |
| $y$ | $-\frac{1}{32}$ | $-\frac{1}{16}$ | $-\frac{1}{8}$ | $-\frac{1}{4}$ | $-\frac{1}{2}$ | -1 | -2 | -4 | -8 | -16 | -32 | -1024 |

- b No. For  $a > 0$  the expression  $a^x$  is greater than zero for all real  $x$ . So,  $-(a^x)$  will be less than or equal to zero.
- c Decreasing
- d As  $x$  increases, the function decreases at a faster and faster rate.
- e All real  $x$ .
- f  $y < 0$

17



- a Positive
- b 0
- c  $y = 0$  is a horizontal asymptote of the curve.

18

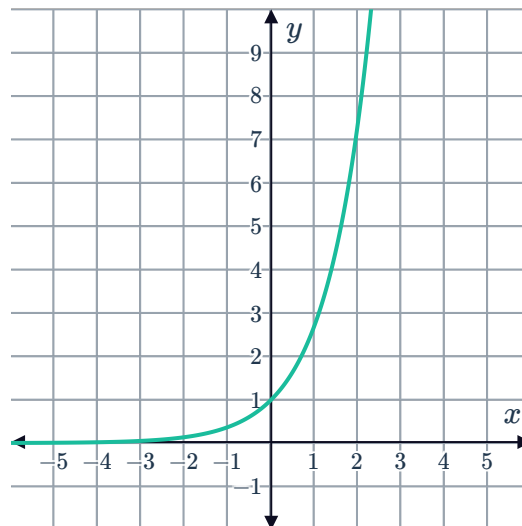
- a  $(0, 1)$
- b No
- c All real  $x$
- d  $y > 0$
- e  $y = 128$
- f  $x = 8$

19

a

|        |      |      |      |      |      |      |       |
|--------|------|------|------|------|------|------|-------|
| $x$    | -3   | -2   | -1   | 0    | 1    | 2    | 3     |
| $f(x)$ | 0.05 | 0.14 | 0.37 | 1.00 | 2.72 | 7.39 | 20.09 |

b



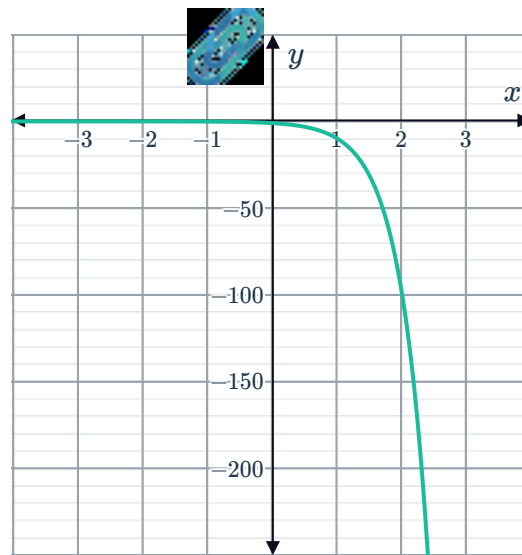
20

a

|     |                  |                 |    |     |      |       |
|-----|------------------|-----------------|----|-----|------|-------|
| $x$ | -2               | -1              | 0  | 1   | 2    | 3     |
| $y$ | $-\frac{1}{100}$ | $-\frac{1}{10}$ | -1 | -10 | -100 | -1000 |

- b Yes. For any value of  $x$ ,  $10^x$  is always positive. Therefore,  $-10^x$  is always negative.

c



21

a No. For  $a > 0$  the expression  $a^x$  is greater than zero for all real  $x$ .

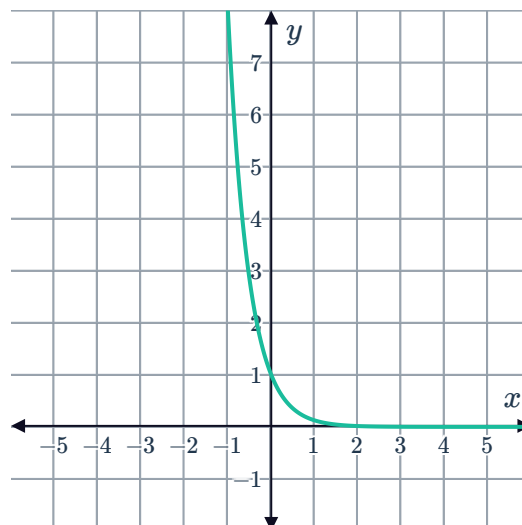
b 0

c  $\infty$

d 1

e 0

f



22

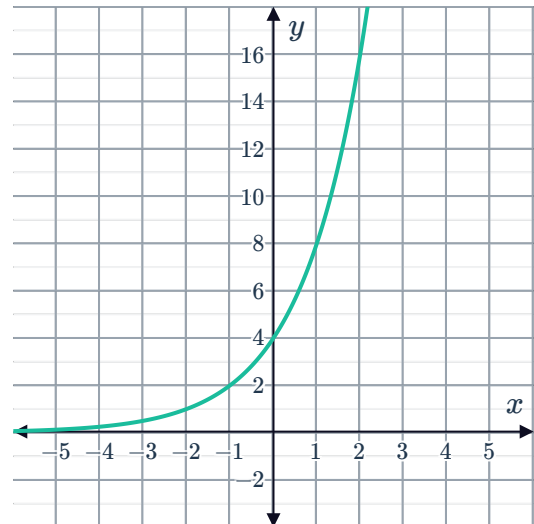
a  $(0, 4)$

c  $\infty$



b No

d



23

a 2

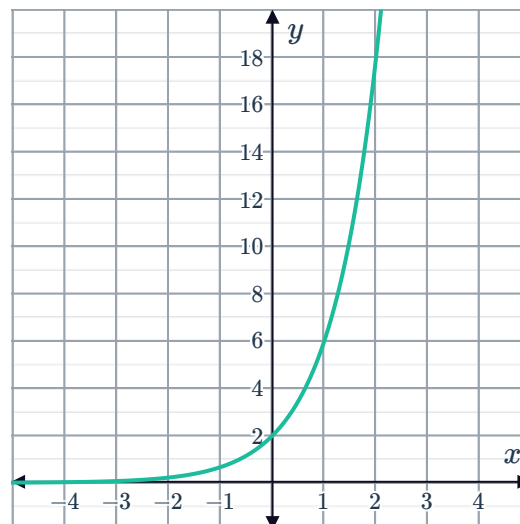
b No. For  $a > 0$  the expression  $a^x$  is greater than zero for all real  $x$ . So,  $2 \times (a^x)$  will be greater than or equal to zero.

c All real  $x$ .

d  $y > 0$

e  $\infty$

f



24

a Graph  $R$

b For  $x < 0$ , the graph of  $y = 6^x$  is below the graph of  $y = 4^x$ . This is because negative values of  $x$  result in fractional function values, and for any negative value of  $x$ ,  $6^x$  will result in a smaller value than  $4^x$ .

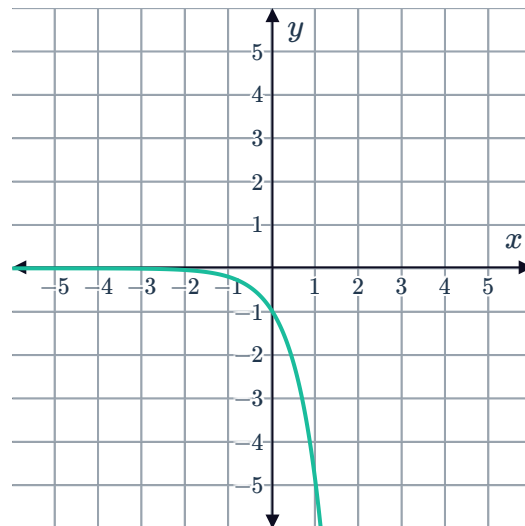
- 25 a** For  $x > 0$ , the graph of  $y = e^x$  will lie above the graph of  $y = (2)^x$  and below the graph of  $y = (3)^x$ .
- b** For  $x < 0$ , the graph of  $y = e^x$  will lie above the graph of  $y = (3)^x$  and below the graph of  $y = (2)^x$ .

## Transformations of exponential functions

**26**

- a** A reflection over the  $x$ -axis

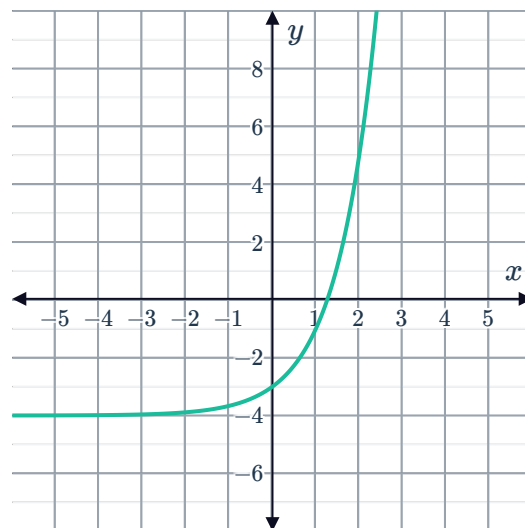
**b**



**27**

- a** Vertical translation

**b**

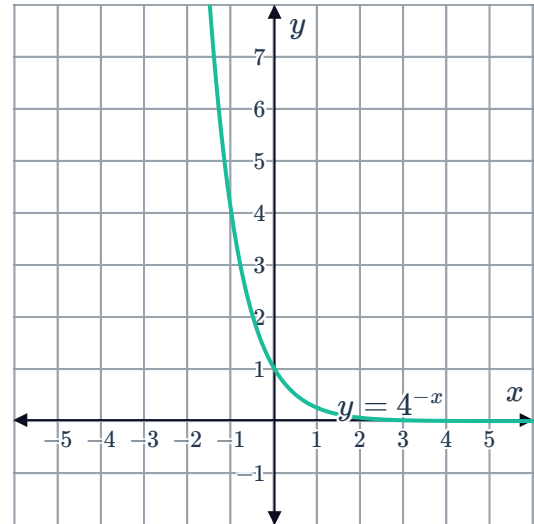


28

a Reflection over the  $y$ -axis.



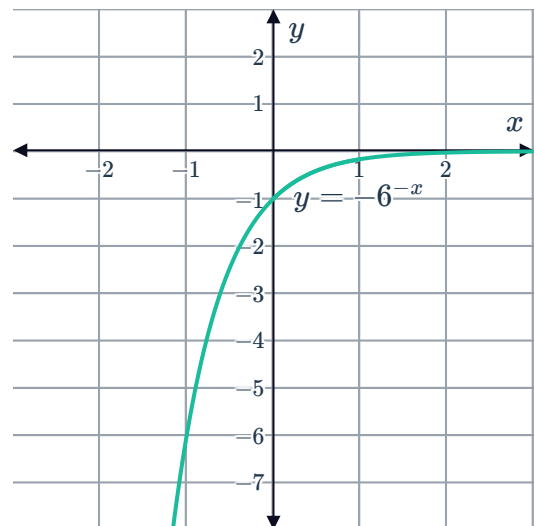
b



29

a Reflection over the  $x$ -axis and  $y$ -axis.

b



30

a  $y = 0$

b  $y = 2$

c 1

d All real  $x$ .

e  $y < 2$

31

a 1

b -1

c  $y = 0$

d  $y = -2$

32



a

i 4 units to the left

ii 2 units

iii 1

b

i 1 units to the left

ii 1 units down

iii  $\frac{1}{3}$ 

c

i  $\frac{1}{5}$  units to the right

ii 2 units

iii 5

33

a  $y = 2^x - 7$

b  $y = 11^{x+10}$

34

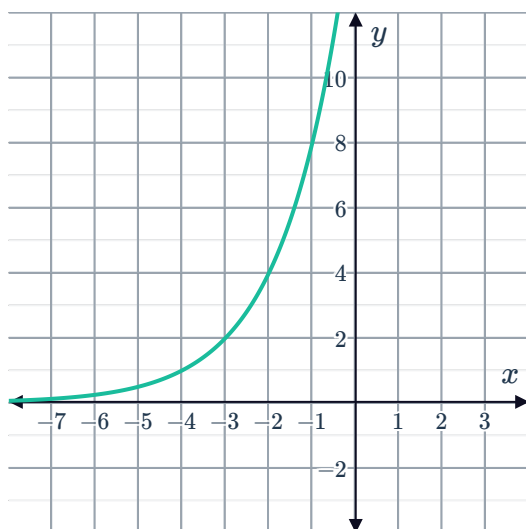
a  $y = 7^{x-3} - 5$

b  $y = -7^{x+3} + 5$

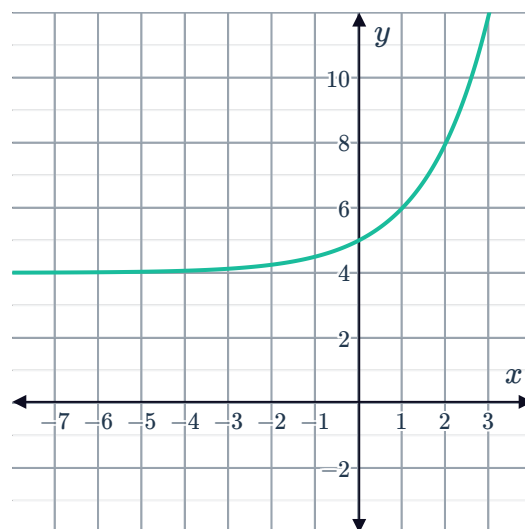
c  $y = -7^{x+3} - 5$

35

a



b





36

a

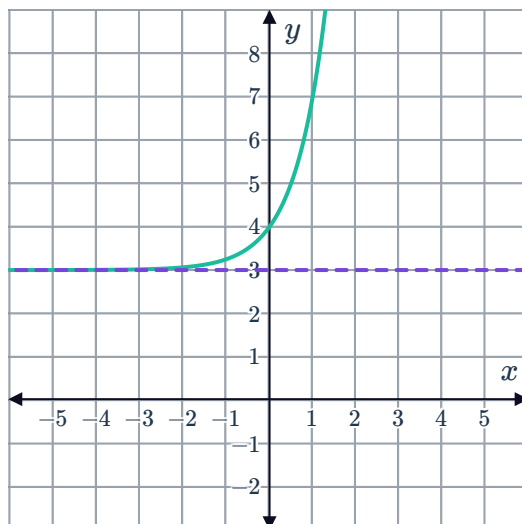
i  $4$

iii  $y > 3$



ii All real  $x$

iv



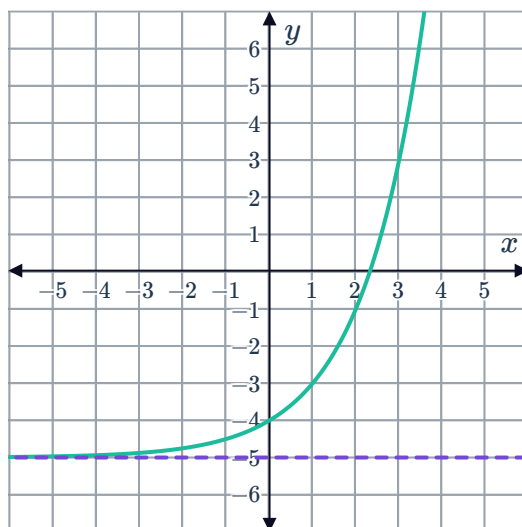
b

i  $-4$

iii  $y > -5$

ii All real  $x$

iv



c

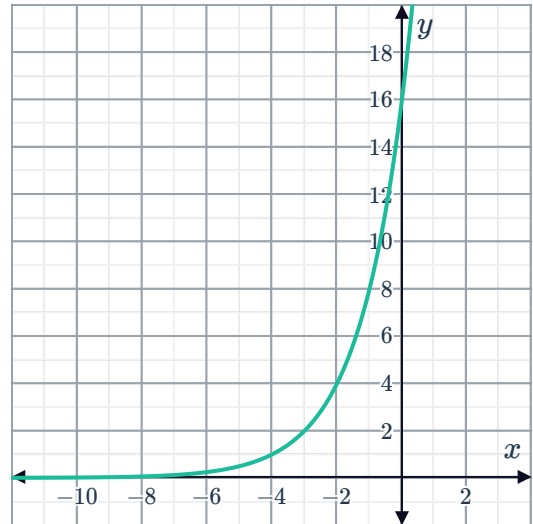
i  $16$

iii  $y > 0$



ii All real  $x$

iv



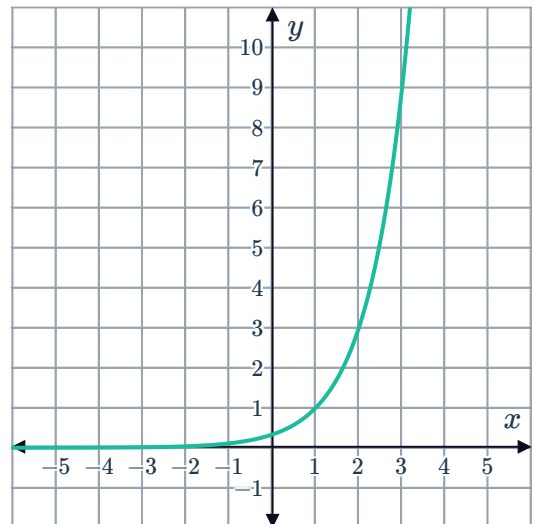
d

i  $\frac{1}{3}$

iii  $y > 0$

ii All real  $x$

iv



e

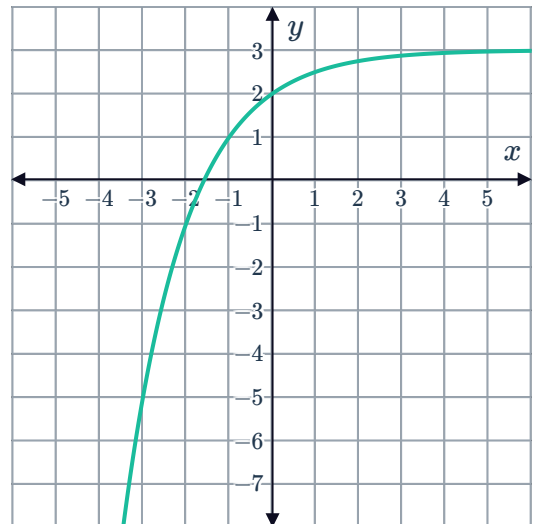
i  $2$

iii  $y < 3$



ii All real  $x$

iv



37

a  $y = 0$

b  $y = 2$

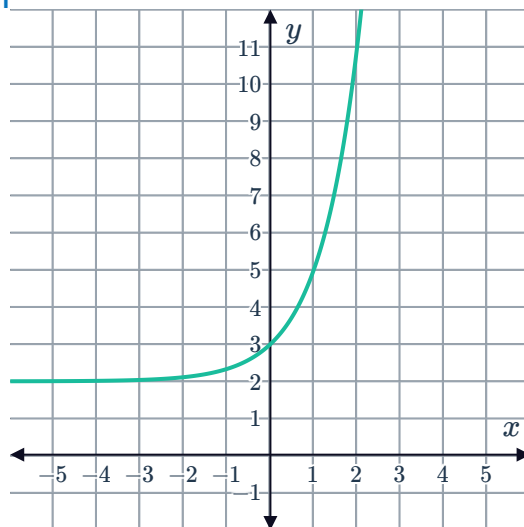
c  $1$

38

a i  $(0, 3)$

ii  $y = 2$

iii



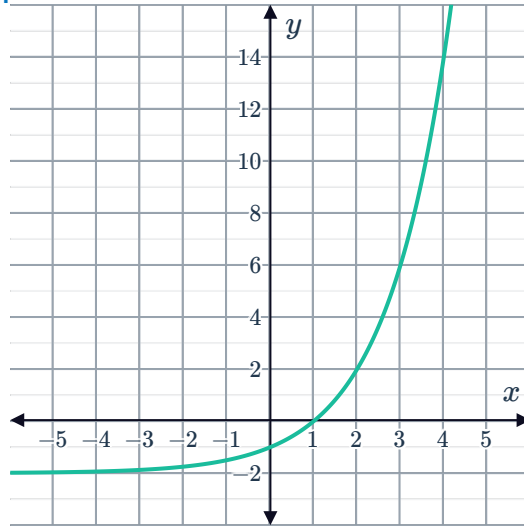
b

i  $(0, -1)$

ii  $y = -2$



iii

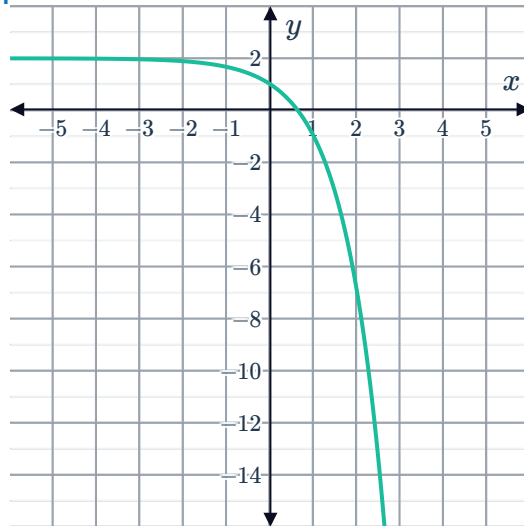


c

i  $(0, 1)$

ii  $y = 2$

iii

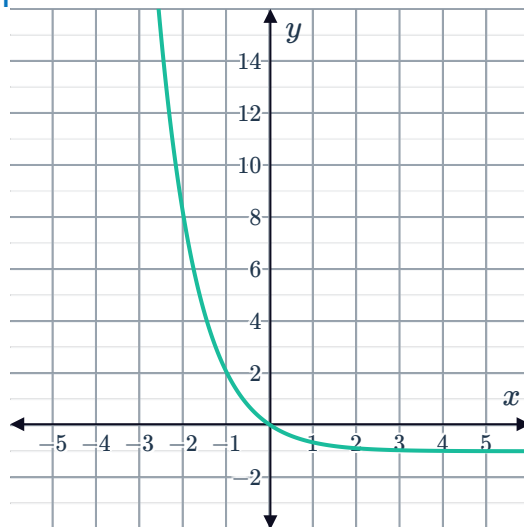


d

i  $(0, 0)$

ii  $y = -1$

iii



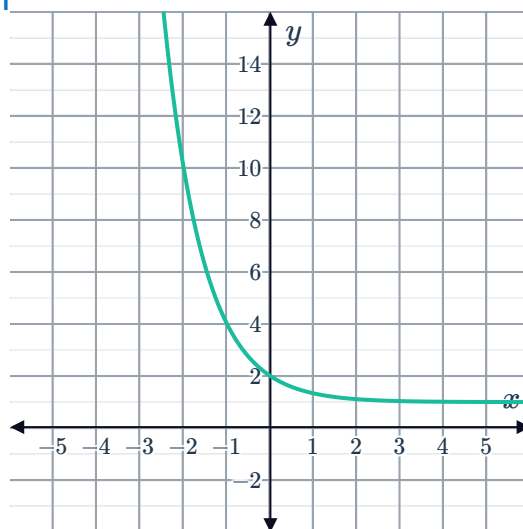
e

i  $(0, 2)$

ii  $y = 1$



iii



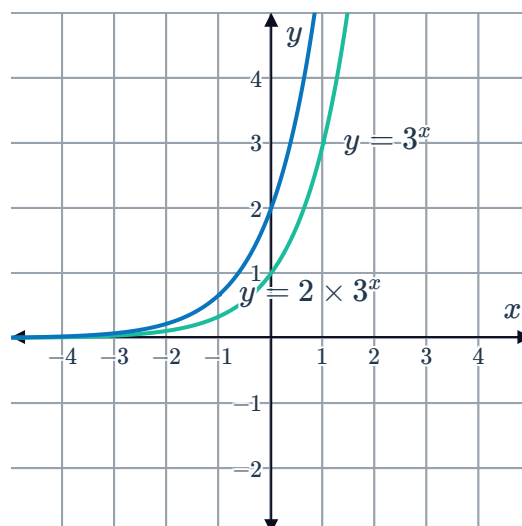
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a

| Point on original graph        | Point on new graph             |
|--------------------------------|--------------------------------|
| $\left(-1, \frac{1}{3}\right)$ | $\left(-1, \frac{2}{3}\right)$ |
| $(0, 1)$                       | $(0, 2)$                       |
| $(1, 3)$                       | $(1, 6)$                       |
| $(2, 9)$                       | $(2, 18)$                      |

b  $y = 2 \times 3^x$

c



d For all values of  $x$ ,  $y = 2 \times 3^x$  lies above  $y = 3^x$ .

40

a

- i No    ii No    iii No

c

- i Yes    ii Yes    iii Yes



b

- i No    ii Yes    iii Yes

d

- i No    ii Yes    iii Yes

## Exponential equations

41

a  $y = 2^x$

b  $y = 2^{-x}$

c  $y = 9^x$

d  $a = 5$

42

a

|     |   |   |   |    |
|-----|---|---|---|----|
| $x$ | 1 | 2 | 3 | 4  |
| $y$ | 2 | 4 | 8 | 16 |

b 2

c  $y = 2^x$

43

a 10

b  $y = 10^x$

c 10 000 000 000

44

a  $a = 5$

b  $b = 4$

c  $y = \frac{5}{16}$